Error Detection and Correction

Introduction

During data transmission, the received data may be different from the original data due to **noise, interference, or signal problems**. This is called an **error**.

Error Detection identifies whether an error has occurred, while **Error Correction** identifies and corrects the erroneous data.

---

1. Types of Errors

There are mainly **two types of errors** in data communication:

A. Single-Bit Error

A **single-bit error** occurs when only **one bit** of the transmitted data changes.

Sent: 10110101

Received: 10100101

↑

Changed bit

Only one bit is affected.

B. Burst Error

A **burst error** occurs when **two or more bits** in a data unit are affected.

Sent: 101101001101

Received: 101011111101

↑↑↑

Multiple errors

Burst errors are more common in data communication systems.

---

2. Error Detection

Error detection is the process of determining whether the received data contains an error.

Extra bits, called **redundant bits**, are added to the original data before transmission.

At the receiver, these bits are checked to determine whether an error occurred.

Common Error Detection Methods

- **VRC (Vertical Redundancy Check)**
- **LRC (Longitudinal Redundancy Check)**
- **CRC (Cyclic Redundancy Check)**
- **Checksum**

---

3. VRC (Vertical Redundancy Check)

VRC is also called **Parity Check**.

In VRC, an extra **parity bit** is added to each data unit to make the number of 1s either **even or odd**.

Example – Even Parity

Data: 1011001

Number of 1s = 4

Parity bit = 0

Transmitted: 10110010

The total number of 1s remains even.

Types

- **Even Parity:** Total number of 1s is even.
- **Odd Parity:** Total number of 1s is odd.

Advantages

- Simple to implement.
- Requires only one extra bit.
- Low overhead.

Limitation

- Cannot reliably detect errors involving an **even number of bit changes**.

---

4. LRC (Longitudinal Redundancy Check)

LRC is also called **two-dimensional parity**.

In LRC, data is arranged in **rows and columns**, and parity bits are calculated for each column.

Example

Data

↓

1 0 1 1

1 1 0 0

0 1 1 0

----

0 0 0 1 ← LRC

The calculated parity bits are added as an additional row.

Advantages

- Better than VRC for detecting errors.
- Can detect many burst errors.
- Simple to implement.

Limitation

- Requires more redundant bits than VRC.
- Some combinations of errors may still go undetected.

---

5. CRC (Cyclic Redundancy Check)

CRC is a powerful and widely used error-detection technique.

It treats the data as a **binary number/polynomial** and performs **binary division** using a generator polynomial.

Working

1. The sender selects a **generator polynomial**.
2. Zeros are added to the data.
3. Binary division is performed using **XOR operation**.

4. The remainder is called the **CRC**.
5. The CRC is added to the original data.
6. The receiver performs the same division.
7. If the remainder is **zero**, the data is assumed to be error-free; otherwise, an error is detected.

Advantages

- Very effective for detecting burst errors.
- More reliable than simple parity methods.
- Widely used in computer networks.

Examples

- Ethernet
- Data storage systems
- Communication protocols

Remember:
CRC = Binary Division + Remainder

---

6. Checksum

A **checksum** is an error-detection method in which data is divided into fixed-size sections and these sections are added together.

The final result is called the **checksum**.

Working

1. Divide data into fixed-size blocks.
2. Add all the blocks using binary addition.
3. Take the complement of the result.
4. The result is the checksum.
5. The checksum is sent along with the data.
6. The receiver performs the same calculation.
7. If the calculated result matches the expected value, the data is considered error-free.

Advantages

- Simple to implement.
- Requires less computational complexity.
- Commonly used in networking protocols.

Limitation

- Less powerful than CRC for detecting certain errors.

---

7. Error Correction

Error correction is the process of **detecting an error and correcting the erroneous data** so that the original data can be recovered.

There are two common approaches:

A. Forward Error Correction (FEC)

- The sender adds extra information to the transmitted data.
- The receiver uses this information to detect and correct errors.
- No retransmission is required.

Example: Hamming Code

B. Retransmission

- The receiver detects an error.
- It requests the sender to **send the data again**.
- The incorrect data is replaced by the retransmitted data.

Example: ARQ (Automatic Repeat reQuest)

Quick Revision Table

Method	Main Idea
VRC	Adds a parity bit
LRC	Uses parity across rows/columns
CRC	Uses binary division and remainder

Method	Main Idea
Checksum	Adds data blocks and checks the result
Error Correction	Detects and corrects/retransmits erroneous data

⭐ Important for Exam

VRC → Parity Bit
LRC → Two-dimensional Parity
CRC → Binary Division
Checksum → Addition of Data Blocks
Error Correction → Correct Error / Retransmit Data

‘

Error Detection

Definition

Error Detection is the process of determining whether the data received by the receiver is **correct or contains errors**.

During data transmission, noise or interference may change one or more bits. To detect these errors, the sender adds some **extra redundant bits** to the original data.

Basic Process

```
Sender              Receiver
 │                   │
 │ Data + Redundant Bits     │
 ├──────────────────────────────────→│
 │                   │
 │          Check for Error
 │                   │
 │        Error? → Yes / No
```

Working

1. The sender adds **redundant bits** to the original data.
2. The data is transmitted through the communication channel.
3. The receiver receives the data along with the redundant bits.
4. The receiver performs a calculation or check on the received data.
5. If the result is incorrect, an **error is detected**.
6. If no error is detected, the data is accepted.

Common Error Detection Methods

1. VRC (Vertical Redundancy Check)

- Also called **Parity Check**.
- Adds one parity bit to the data.
- Can use even or odd parity.

2. LRC (Longitudinal Redundancy Check)

- Also called **two-dimensional parity**.
- Checks data in rows and columns.
- Provides better error detection than simple parity.

3. CRC (Cyclic Redundancy Check)

- Uses **binary division**.
- The remainder is added to the data.
- Very effective for detecting burst errors.

4. Checksum

- Data is divided into fixed-size blocks.
- Blocks are added together.
- The result is used to detect errors.

Advantages

- Detects errors during data transmission.
- Improves the reliability of communication.
- Helps prevent incorrect data from being accepted.
- CRC and other methods can detect many types of errors.

Important Point

Error Detection does not normally correct the error. It only determines that an error has occurred. After detection, the system may use **retransmission or an error-correction technique** to recover the correct data.

⭐ **Exam Definition**

Error detection is the technique of identifying errors in transmitted data by adding redundant information and checking the received data at the destination.

Remember:
Error Detection = Add Redundant Bits → Check Received Data → Detect Error

Bilkul bhai. **Mid-level detail** mein de raha hoon — exam mein directly likh sakta hai, aur points bhi enough hain.

Error Detection and Correction Techniques

1. Vertical Redundancy Check (VRC)

Definition

VRC (Vertical Redundancy Check) is an error detection technique in which an extra **parity bit** is added to each data unit.

The parity bit makes the total number of 1s either **even or odd**.

Types

- **Even Parity:** Total number of 1s is made even.
- **Odd Parity:** Total number of 1s is made odd.

Example

For even parity:

Data: 1011001

Number of 1s = 4

Parity bit = 0

Transmitted: 10110010

Advantages

- Simple and easy to implement.
- Requires only one extra bit.
- Suitable for detecting single-bit errors.

Limitation

- Cannot detect all errors, especially when an **even number of bits** are changed.

---

2. Longitudinal Redundancy Check (LRC)

Definition

LRC (Longitudinal Redundancy Check) is an error detection technique in which data is arranged in **rows and columns**, and parity is calculated for each column.

It is also called **two-dimensional parity**.

Working

1. Data is divided into blocks.
2. The blocks are arranged in rows.
3. Parity bits are calculated column-wise.
4. These parity bits form the **LRC**.
5. LRC is transmitted along with the original data.
6. The receiver checks the parity to detect errors.

Example

Data

↓

1 0 1 1

1 1 0 0

0 1 1 0

----

0 0 0 1 ← LRC

Advantages

- Better error detection than VRC.
- Can detect many **burst errors**.
- Simple to implement.

Limitation

- Requires more redundant information than VRC.
- Some multiple-bit errors may remain undetected.

---

3. Cyclic Redundancy Check (CRC)

Definition

CRC (Cyclic Redundancy Check) is a powerful error detection technique that uses **binary division** to detect errors in transmitted data.

It is widely used in computer networks.

Working

1. The sender selects a **generator polynomial**.
2. Extra zeros are added to the data.
3. Binary division is performed using **XOR operation**.
4. The remainder obtained is called the **CRC**.
5. The CRC is added to the original data.
6. The receiver performs the same calculation.
7. If the remainder is zero, the data is considered error-free; otherwise, an error is detected.

Advantages

- Very effective for detecting **burst errors**.
- More reliable than simple parity checking.
- Widely used in networking.

Example

Ethernet uses CRC for error detection.

Remember:
CRC = Binary Division → Remainder → Error Detection

---

4. Checksum

Definition

A **checksum** is an error detection technique in which data is divided into fixed-size blocks and the blocks are **added together** to generate a checksum.

Working

1. Data is divided into equal-sized blocks.
2. All blocks are added using binary addition.
3. The complement of the result is taken as the **checksum**.

4. The checksum is sent along with the data.
5. The receiver performs the same calculation.
6. If the result matches, the data is considered correct.

Advantages

- Simple to understand and implement.
- Requires less computational effort.
- Used in several networking protocols.

Limitation

- Less powerful than CRC for detecting certain errors.

Remember:
Checksum = Divide → Add → Complement → Check

---

5. Error Correction

Definition

Error Correction is the process of **detecting and correcting errors** in received data so that the original data can be recovered.

Unlike error detection, error correction can determine the incorrect data and correct it.

Methods of Error Correction

A. Forward Error Correction (FEC)

- Extra redundant bits are added at the sender.
- The receiver uses these bits to **detect and correct errors**.
- No retransmission is required.
- Example: **Hamming Code**.

B. Retransmission

- The receiver detects an error.
- It requests the sender to **send the data again**.
- The incorrect data is replaced by the correct retransmitted data.
- Example: **ARQ (Automatic Repeat reQuest)**.

Advantages

- Improves reliability of communication.
- Corrects errors in transmitted data.
- Reduces the chance of incorrect data being used.

Quick Revision Table

Technique	Main Principle
VRC	Parity bit
LRC	Row/Column parity
CRC	Binary division
Checksum	Addition of data blocks
Error Correction	Detect and correct/retransmit

⭐ **Easy Memory Trick**

VRC → Parity
LRC → Columns
CRC → Division
Checksum → Addition
Correction → Fix Error

Data Link Layer Protocols

The **Data Link Layer** is responsible for reliable communication between two directly connected devices. It provides functions such as **framing, error control, flow control and access control**.

---

1. Framing

Definition

Framing is the process of dividing a continuous stream of bits into smaller units called **frames** at the Data Link Layer.

A frame contains the actual data along with control information.

Basic Structure

```
┌────────┬──────────┬─────────────┬─────────┐
│ Header │  Data  │  Trailer │  FCS  │
└────────┴──────────┴─────────────┴─────────┘
        Frame
```

Functions of Framing

- Divides data into manageable frames.
- Identifies the **start and end** of data.
- Provides addressing information.
- Helps in error detection.
- Provides synchronization between sender and receiver.

Methods of Framing

- Character/Byte stuffing
- Bit stuffing
- Character count
- Physical layer coding violations

---

2. HDLC

Definition

HDLC (High-Level Data Link Control) is a **bit-oriented Data Link Layer protocol** used for reliable communication between devices.

It provides **framing, flow control and error control**.

HDLC Frame Format

```
┌───────┬──────────┬─────────┬─────────┬───────┬───────┐
│Flag │ Address │Control │ Data │ FCS │ Flag │
└───────┴──────────┴─────────┴─────────┴───────┴───────┘
```

Important Fields

- **Flag:** Marks the beginning and end of a frame.
- **Address:** Identifies the receiving station.
- **Control:** Contains control information.
- **Data:** Contains the actual user data.
- **FCS:** Used for error detection.

Types of HDLC Frames

1. **I-Frame (Information Frame)** – Carries user data.
2. **S-Frame (Supervisory Frame)** – Used for flow and error control.
3. **U-Frame (Unnumbered Frame)** – Used for link management and control.

Features

- Bit-oriented protocol.
- Uses bit stuffing.
- Provides error and flow control.
- Uses CRC/FCS for error detection.
- Provides reliable data transmission.

---

3. ARQ

Definition

ARQ (Automatic Repeat reQuest) is an error-control technique in which the receiver requests the sender to **retransmit data when an error is detected**.

ARQ generally uses:

- **Acknowledgement (ACK)**
- **Negative Acknowledgement (NAK)** or timeout
- **Retransmission**

There are two important ARQ methods:

A. Stop-and-Wait ARQ

Definition

In **Stop-and-Wait ARQ**, the sender sends **one frame at a time** and waits for an acknowledgement before sending the next frame.

Working

```
Sender                 Receiver
 |                   |
 |──────── Frame 1 ──────────────────→|
 |                   |
 |←────────── ACK ──────────────────────|
 |                   |
 |──────── Frame 2 ──────────────────→|
 |                   |
 |←────────── ACK ──────────────────────|
```

If the frame or ACK is lost, the sender waits for a **timeout** and retransmits the frame.

Advantages

- Simple to implement.
- Easy error control.
- Requires less complex hardware.

Limitation

- **Low efficiency** for long-distance or high-speed networks because the sender must wait after every frame.

B. Sliding Window Protocol

Definition

Sliding Window is a flow and error-control technique that allows the sender to transmit **multiple frames before receiving acknowledgements**.

The sender maintains a **window of frames** that can be sent without waiting.

Working

Window:

```
┌────┬────┬────┬────┐
│ F1 │ F2 │ F3 │ F4 │ → Send
└────┴────┴────┴────┘

     ↓

   ACKs received

     ↓

Window moves forward

┌────┬────┬────┬────┐
│ F3 │ F4 │ F5 │ F6 │
└────┴────┴────┴────┘
```

Advantages

- Higher efficiency than Stop-and-Wait.
- Allows multiple frames to be transmitted.
- Better utilization of the communication channel.
- Suitable for high-speed networks.

Types

- **Go-Back-N ARQ**
- **Selective Repeat ARQ**

---

4. Efficiency

Definition

Efficiency represents how effectively the communication channel is utilized for transmitting useful data.

It depends mainly on:

- Transmission time
- Propagation delay
- Frame size
- Acknowledgement time
- Window size

Stop-and-Wait

In Stop-and-Wait, the sender waits for an ACK after every frame, so the channel may remain **idle for a significant amount of time**.

Therefore:

Stop-and-Wait → Lower Efficiency

Sliding Window

In Sliding Window, multiple frames can be transmitted before waiting for acknowledgements.

Therefore:

Sliding Window → Higher Efficiency

Quick Comparison

Feature	Stop-and-Wait ARQ	Sliding Window
Frames sent at a time	One	Multiple
Waiting	After every frame	After window is used
Efficiency	Low	High
Complexity	Simple	More complex
Suitable for	Small/slow networks	High-speed networks

⭐ Exam Summary

Framing divides data into frames, HDLC provides reliable Data Link Layer communication, ARQ provides error control through retransmission, Stop-and-Wait sends one frame at a time, while Sliding Window allows multiple frames to be sent before acknowledgement.

Functions of TCP/IP Layers

The **TCP/IP protocol suite** is divided into four layers. Each layer performs a specific function in the process of data communication.

1. Network Access Layer

It is the **lowest layer** of the TCP/IP model. It is responsible for transmitting data between devices connected to the same physical network.

Functions:

- Transmits data over the physical medium.
- Performs framing.
- Uses physical/MAC addresses.
- Controls access to the transmission medium.
- Handles communication between directly connected devices.

Examples: Ethernet, Wi-Fi

---

2. Internet Layer

The Internet Layer is responsible for **delivery of packets from source to destination across different networks**.

Functions:

- Provides logical addressing using IP addresses.
- Performs routing.
- Selects the path for packet delivery.
- Forwards packets between networks.
- Handles packet delivery.

Protocols: IP, ICMP, ARP

---

3. Transport Layer

The Transport Layer provides **end-to-end communication** between applications on different devices.

Functions:

- Divides data into smaller segments.
- Provides error control.
- Provides flow control.
- Provides reliable data delivery when required.
- Uses port numbers to identify applications.

Protocols: TCP, UDP

---

4. Application Layer

The Application Layer provides **network services directly to user applications**.

Functions:

- Provides web communication.
- Supports file transfer.
- Supports email communication.
- Provides domain name services.
- Supports remote access.

Protocols: HTTP, FTP, SMTP, DNS, SSH

Transmission Media

Definition

Transmission media is the physical or wireless path through which data signals travel from a **sender to a receiver**.

Transmission media is mainly divided into two types:

```
Transmission Media
   │
   ├── Guided Media
```

│

└── Unguided Media

---

1. Guided Media

Guided media is a type of transmission medium in which signals travel through a **physical path or cable**.

Types of Guided Media

a) Twisted Pair Cable

It consists of **two insulated copper wires twisted together** to reduce interference.

Types:

- UTP – Unshielded Twisted Pair
- STP – Shielded Twisted Pair

Uses: Telephone lines, Ethernet networks.

Advantages:

- Low cost.
- Easy to install.
- Suitable for short-distance communication.

Limitation:

- More affected by noise compared with fiber optic cable.

---

b) Coaxial Cable

A coaxial cable consists of a **central conductor, insulating layer, metallic shield and outer covering**.

Outer Cover

┌──────────────┐

│ Metallic Shield │

│ Insulator │

│ ● Conductor │

└──────────────┘

Uses:

- Cable TV
- Broadband communication
- Older computer networks

Advantages:

- Better protection from interference than twisted pair.
- Can carry signals over longer distances.

---

c) Optical Fiber

Optical fiber uses **light signals** to transmit data through a thin glass or plastic fiber.

Types:

- Single-mode fiber
- Multimode fiber

Advantages:

- Very high bandwidth.
- High transmission speed.
- Low signal loss.
- Not affected by electromagnetic interference.
- Suitable for long-distance communication.

Uses: Internet backbone, telephone networks, cable TV.

---

2. Unguided Media

Unguided media is a wireless transmission medium in which signals travel through **air or free space** without using a physical cable.

Types of Unguided Media

a) Radio Waves

Radio waves are used for **wireless communication** and can travel over relatively long distances.

Uses:

- Radio broadcasting
- Mobile communication
- Wireless networks

b) Microwaves

Microwaves are high-frequency electromagnetic waves used for **point-to-point communication**.

Uses:

- Satellite communication
- Mobile networks
- Long-distance communication

c) Infrared

Infrared communication uses infrared light for **short-distance communication**.

Uses:

- TV remote controls
- Short-range device communication

Guided vs Unguided Media

Guided Media	**Unguided Media**
Uses a physical cable	Does not use a physical cable
Signals travel through a guided path	Signals travel through air/free space
More secure generally	More susceptible to interception
Installation may be difficult	Easier to deploy in many situations
Examples: Twisted pair, coaxial, fiber	Examples: Radio, microwave, infrared

⭐ **Exam Definition**

Transmission media is the communication path through which data signals are transmitted from a sender to a receiver. It is classified into guided media and unguided media.

Remember:
Guided = Cable
Unguided = Wireless